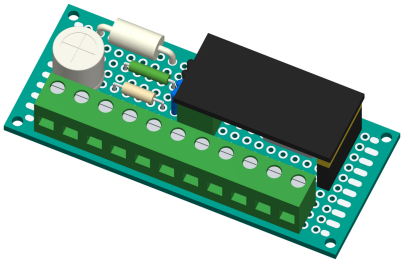
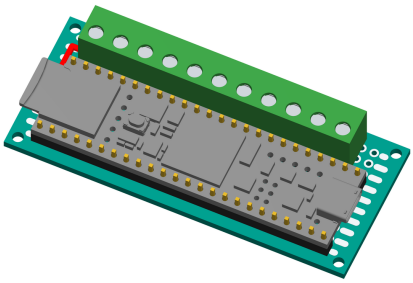
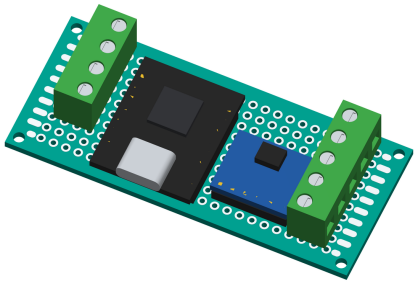
The boards

How they work, how to build them







Designed by Athanasia Nikolova

v1.1 · October 2026

All data come from tests and many real builds, not only from theory.

Teacher only

- **One thing per page**
  One action, one big picture. The part of the step is the only strong colour, the rest is quiet grey. Short sentences, much white space.
- **Colour is never the only clue**
  Every wire on a board drawing also has a number, every board a letter (X, Y, Z), every insert its size. The insert colours stay distinct with colour-vision deficiency (Okabe-Ito palette).
- **Readable**
  All instruction text has a contrast of at least 4.5 : 1 to the page (WCAG 2.x, AA).
- **Print it right**
  In colour, at 100 % - not 'fit to page'. The Assembly guide draws every screw at real size: the student lays the screw on the page to check it.
- **Pace**
  One page, then the action, then the next page. For a student who tires quickly: one chapter per session. The check pages are good stopping points.

✋ Safety first

- **Grown-up nearby**
  Work with your teacher or mentor in the room.

- **Hot!**
  The iron is hot. Put it back in its stand every time.

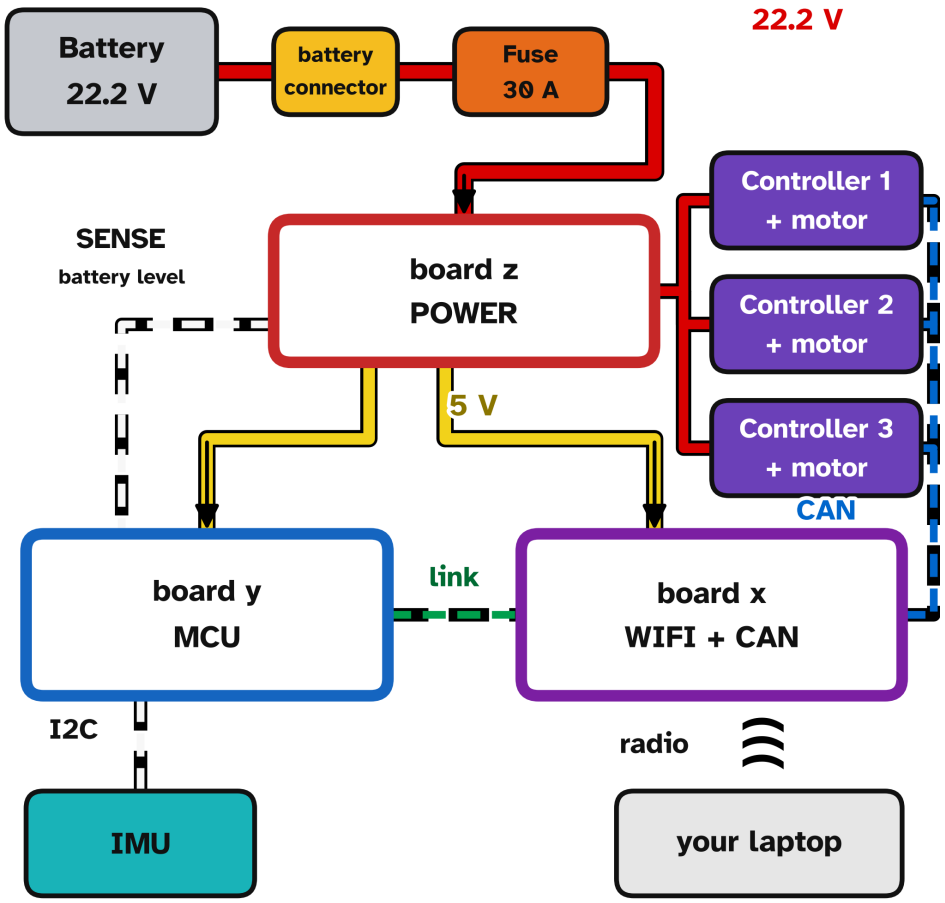
- **Fresh air**
  Solder smoke is not good to breathe. Open a window.

- **Eyes**
  Wear glasses when you cut wire legs - they fly.

- **Power last**
  Connect the battery only after the check pages are done.

The big picture

Thick lines carry power. Dashed lines carry messages.



Power: 22.2 V and 5 V

Battery

6 cells x 3.7 V = 22.2 V (25.2 V when full).

Fuse 30 A

If too much current flows, it melts and cuts the power.

board z

Gives 22.2 V to the 3 motor controllers.

DC-DC

Turns 22.2 V into 5 V for boards y and x.

MCU

Makes its own 3.3 V from the 5 V.

How much current?

Balancing still: about 0.9 A. One wheel at full power: 10 A. All three at the worst moment: 30 A - that is why the fuse is 30 A.

Messages: who talks to whom

MCU

the brain. It decides everything.

IMU -> MCU

how the cube is tilted (I2C wires).

SENSE -> MCU

how full the battery is.

MCU -> CAN

tells each motor controller how hard to turn.

MCU <-> Wi-Fi board

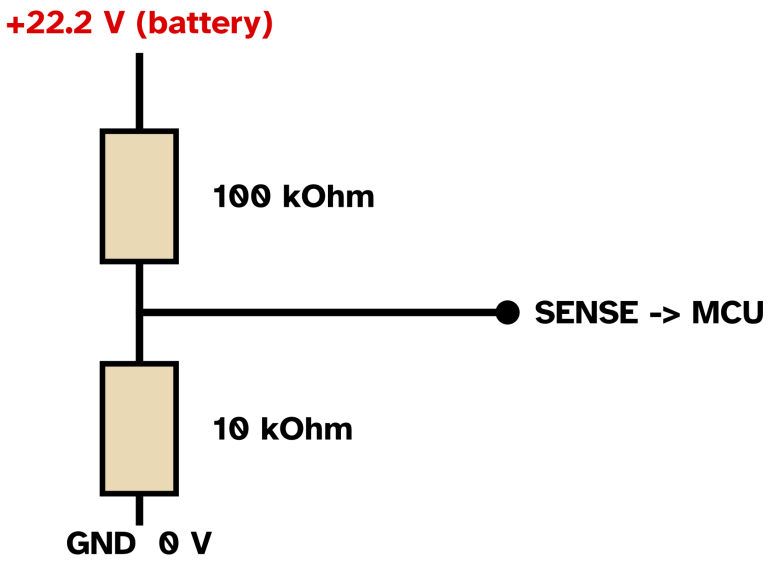
the link cable between boards y and x.

Wi-Fi board <-> laptop

by radio: you see the data, you send commands.

How the MCU reads the battery

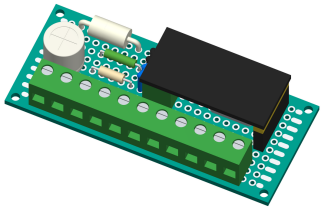
The MCU can read at most 3.3 V. Two resistors make the battery voltage 11 times smaller.



25.2 V x 10 / 110 = 2.29 V	full
22.2 V x 10 / 110 = 2.02 V	normal
19.8 V x 10 / 110 = 1.80 V	empty - stop!

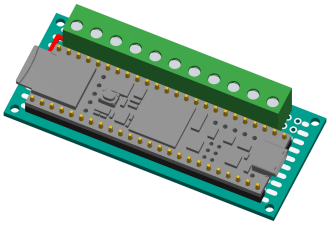
Meet the three boards

Each board has its own colour on every page.



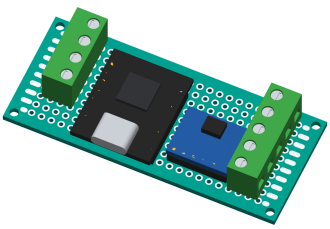
board z · POWER

Takes the battery (22.2 V), feeds the 3 motor controllers and makes 5 V for the other two boards.



board y · MCU

The brain: the MCU board reads the IMU and the battery voltage and decides how the wheels must turn.

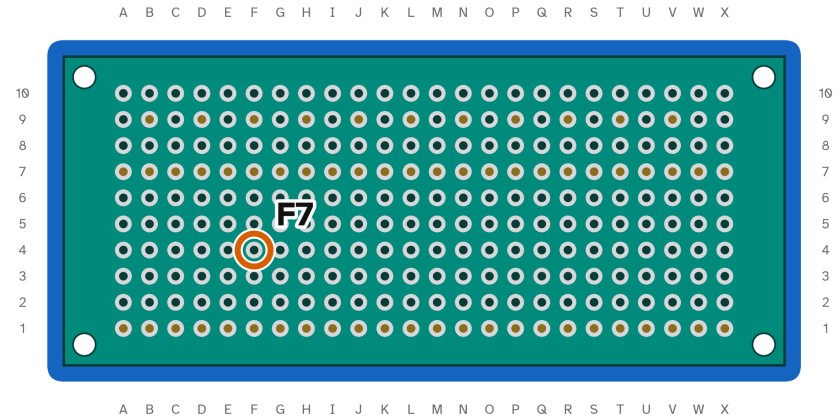


board x · WIRELESS + CAN

Talks to your laptop by radio (Wi-Fi board) and to the 3 motor controllers over the CAN bus (CAN board).

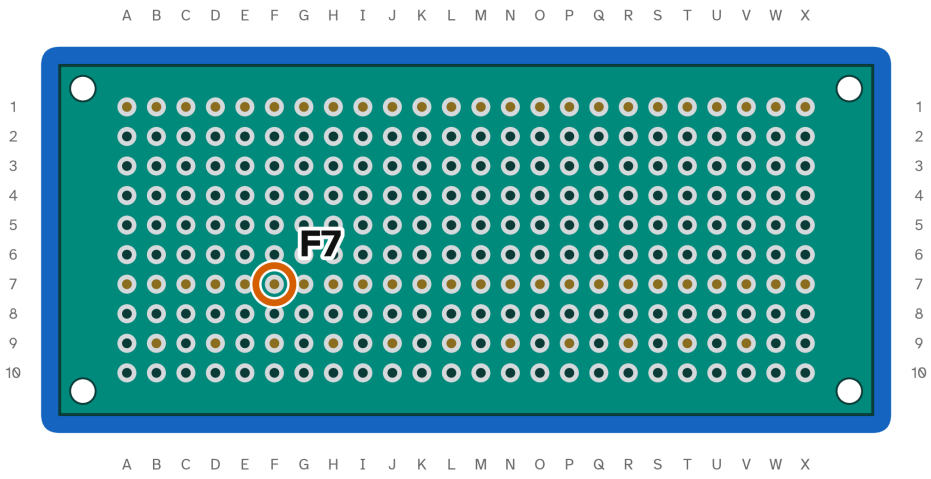
Find a hole

Letter = row (A … X). Number = column (1 … 10).



Hole F7: row F, column 7.

Turned over, the letters stay in place. The numbers go the other way:



Wire colours

Every wire is BARE tinned copper. The colours are only in the drawings - so they are easy to read.

	**red**	+22V battery power
	**black**	GND ground, 0 V
	**yellow**	+5V
	**white**	message
	**blue**	message
	**green**	message
	**copper**	bus bar (board z), soldered all along

Each wire has a number, outside the board, with a thin line to it.

A dot on a wire = a solder joint: its two ends, and every pin it is soldered to on the way.

The order of work

The same on every board.

1 **Parts**

Put each part in its holes, on the top side.

2 **Solder**

Turn the board over. Solder every pin.

3 **Power wires**

Bus bars, red, black, yellow - first, they lie lowest.

4 **Signal wires**

White, blue, green - after the power.

5 **Check**

Multimeter: beep where it must, silence where it must not.

All wires go on the BOTTOM side (the copper side). All of them are bare wire: they never touch each other, nor a pin of another wire.

Soldering school

Every joint is made the same way: heat, solder, away.

- Solder melts only on hot metal. Heat the metal, not the solder.
- A good joint is a small shiny volcano round the pin.
- Practise 10 joints on a spare board before the real one.
- The iron at 350 °C. Wire takes the heat away: hold the tip a little longer, never more than 3-4 s on one joint.

Materials for good joints

- Solder: tin-lead 60/40 or 63/37, 0.6-0.8 mm, with a flux core. Wash your hands after.
- Optional - flux: no-clean flux gel in a syringe. A tiny drop on the pad and on the wire before each joint: the solder flows better and the joint shines.
- Optional - contact cleaner: isopropyl alcohol (IPA, 99 %) in a dropper bottle. Clean the pads and the old copper before you solder; clean the flux off at the end.
- Wire: bare tinned copper, solid, 0.6 mm (22 AWG).
- Iron tips: a fine conical one for the pins, a 2 mm chisel one for the wire.
- Brass wool to clean the tip; tip tinner when the tip turns black.
- Desoldering pump and wick, for mistakes.
- Flat-nose pliers, side cutter, tweezers.
- Helping hands with a magnifier, or a board holder - and a lamp.
- An old toothbrush, with the alcohol, to scrub the board clean at the end.
- Glasses and a small fan: the smoke goes away from your face.

Bare wire, step by step

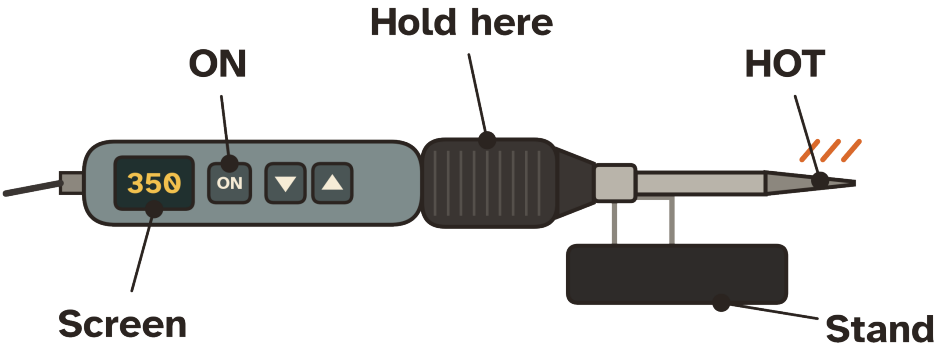
1. Straighten it: hold both ends with pliers and pull.
2. Tin it: a little flux, then run the hot tip with solder along it.
3. Bend every corner with the flat-nose pliers, right over the centre of a hole.
4. Lay it flat on the board. Solder both ends first, then every dot on the way.
5. Cut the spare end short, the flat side of the cutter towards the joint.
6. Check: it touches no other wire and no pin of another wire.

Your tools

Set them out before you start.

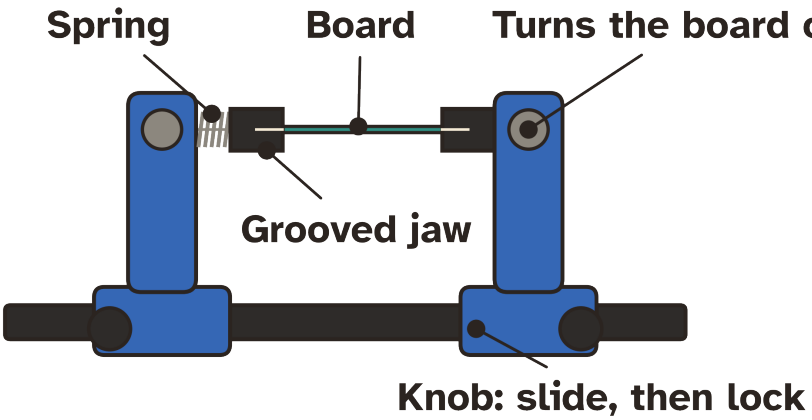
Soldering iron

Conical tip for the pins, 2 mm chisel tip for the wire. Set it to 350 °C. Back in the stand after every joint.



Board holder

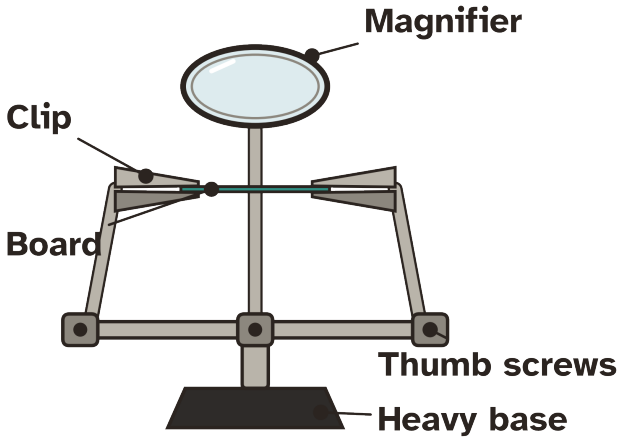
The board sits in the two grooved jaws. Turn it over without taking it out.



Your tools

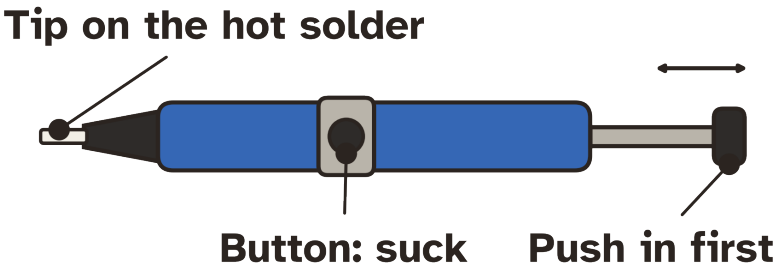
Helping hands

Two clips hold the board. Look through the magnifier at every joint.



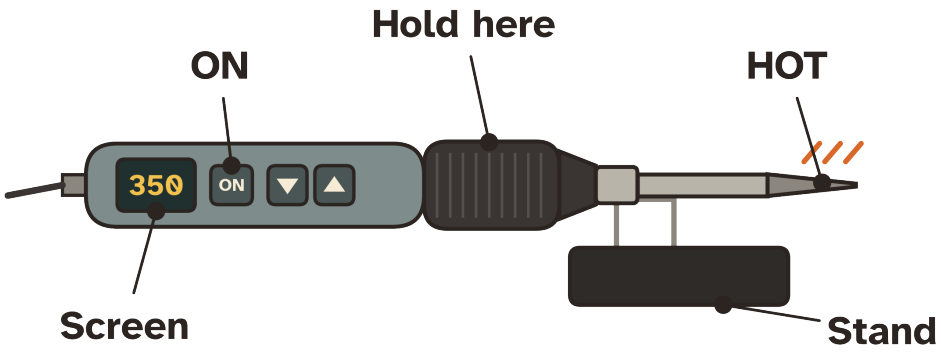
Desoldering pump

For mistakes: push the knob in, heat the solder, tip on it, press the button.



Hold the iron

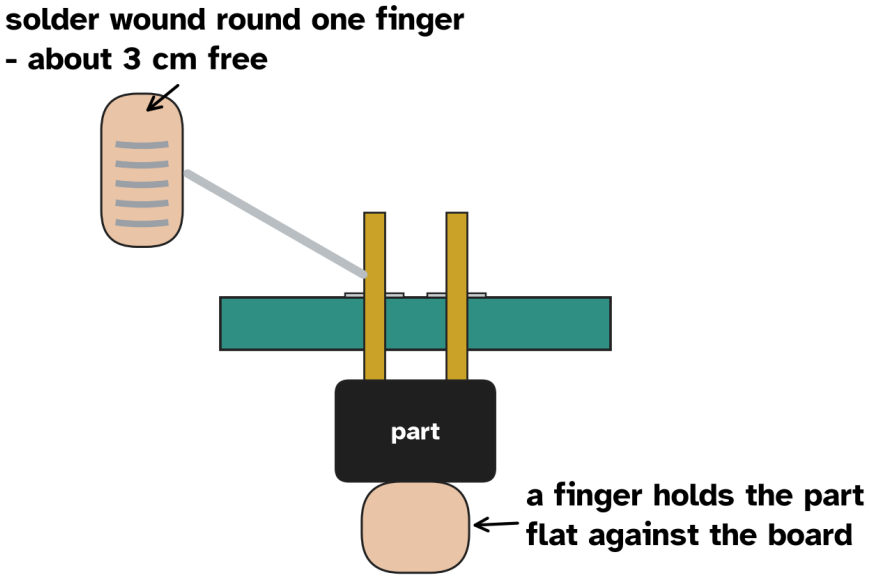
Like a pen - on the grip, never on the metal.



- Hold it on the soft grip, like a pen.
- Rest your hand on the table: the tip stays steady.
- Back in the stand after every joint.
- Clean the tip in the brass wool. A clean tip is shiny.

Hold the solder

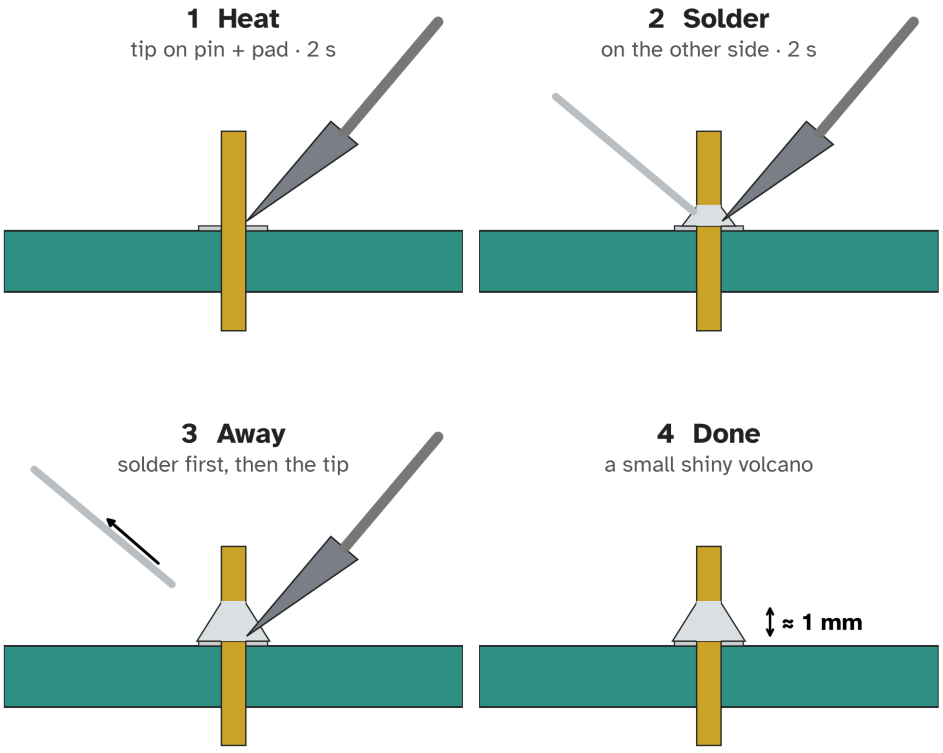
One hand holds the iron. The other hand does two jobs.



- Wind the solder round one finger. Leave about 3 cm free.
- With another finger of the same hand, press the part flat against the board.
- Feed the solder from that hand, a little at a time.

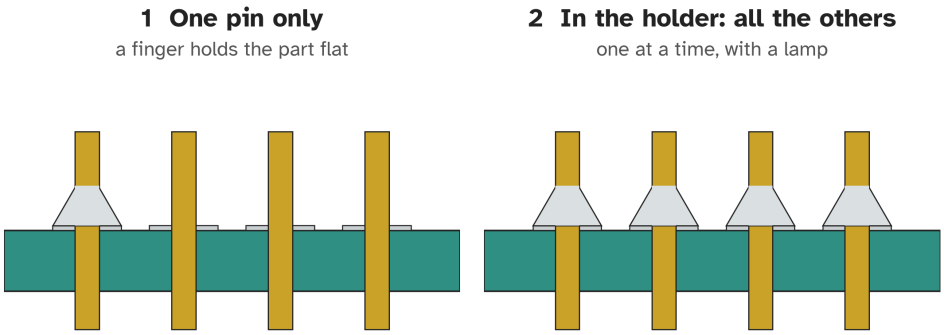
One joint, step by step

1. Tip on the pin AND the pad together. Count 2 s.
2. Touch the solder on the other side of the pin. Count 2 s more. Add as much as it needs: the pad covered and about 1 mm up the pin.
3. Take the solder away first, then the tip.
4. Do not move the part for a few seconds: the solder is setting.



First one pin, then the holder

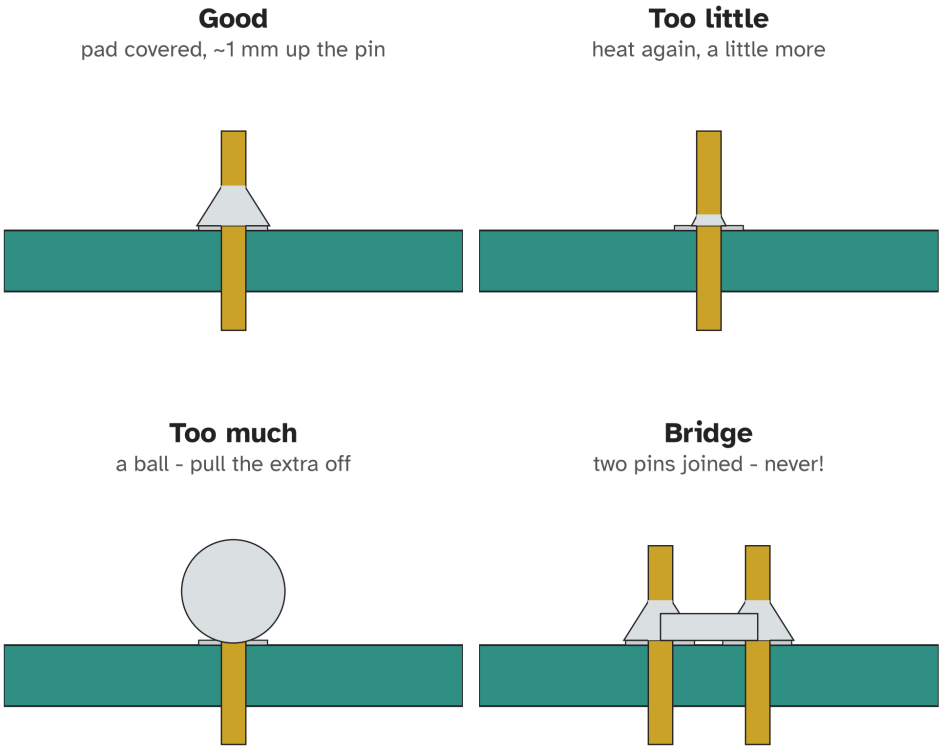
1. Solder ONE pin only, while your finger holds the part flat.
2. Look from the side: is the part flat on the board? If not, heat that pin and press again.
3. Now put the board in the holder (helping hands) and turn on the lamp.
4. Solder every other pin, one at a time, carefully.



Good or not?

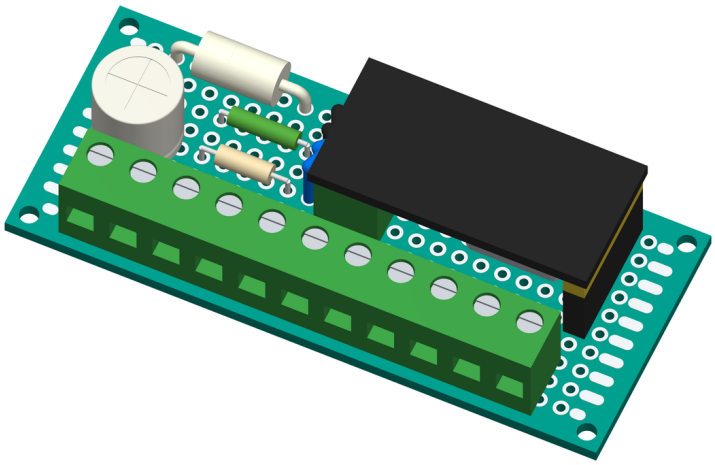
Look at every joint with the lamp.

- Too little: heat again and add a little solder.
- Too much: heat it and pull the extra away with the clean tip.
- Bridge: heat it and pull the tip between the pins, or use the pump or the wick.
- Dull or grainy: heat again for 2 s.



board z · POWER

Takes the battery (22.2 V), feeds the 3 motor controllers and makes 5 V for the other two boards.



10 parts · 18 wires · 2 bus bars

Face z of the box. Find the letter Z on the box wall.

board z · the parts

1x **Terminal block, 6 poles (2 x 5.08 mm 3-pole)**
motors: M1+ M1- M2+ M2- M3+ M3-
holes: B2 … L2

1x **Terminal block, 2 poles (5.08 mm 2-pole)**
battery: + and -
holes: N2 +, P2 -

1x **Terminal block, 3 poles (5.08 mm 3-pole)**
out to y and x: SENSE, GND, +5V
holes: R2, T2, V2

1x **DC-DC 22.2 V -> 5 V module**
turned over (marking up), its pins in a female header in row W; a wire from each of its 4 screw poles straight down
holes: GND W10-W8, VOUT W7-W5; poles GND J9, J8, VIN J7, J6

1x **Pin header, female, 1 x 6, 8.5 mm**
holds the DC-DC
holes: W10 … W5

1x **TVS diode 1.5KE33CA - bidirectional: either way round**
protects +22V from spikes
holes: B10 - H10

1x **Capacitor 220 uF 50 V - polarized**
+ leg (long) in C6, the stripe (-) side in A6
holes: C6 +, A6 -

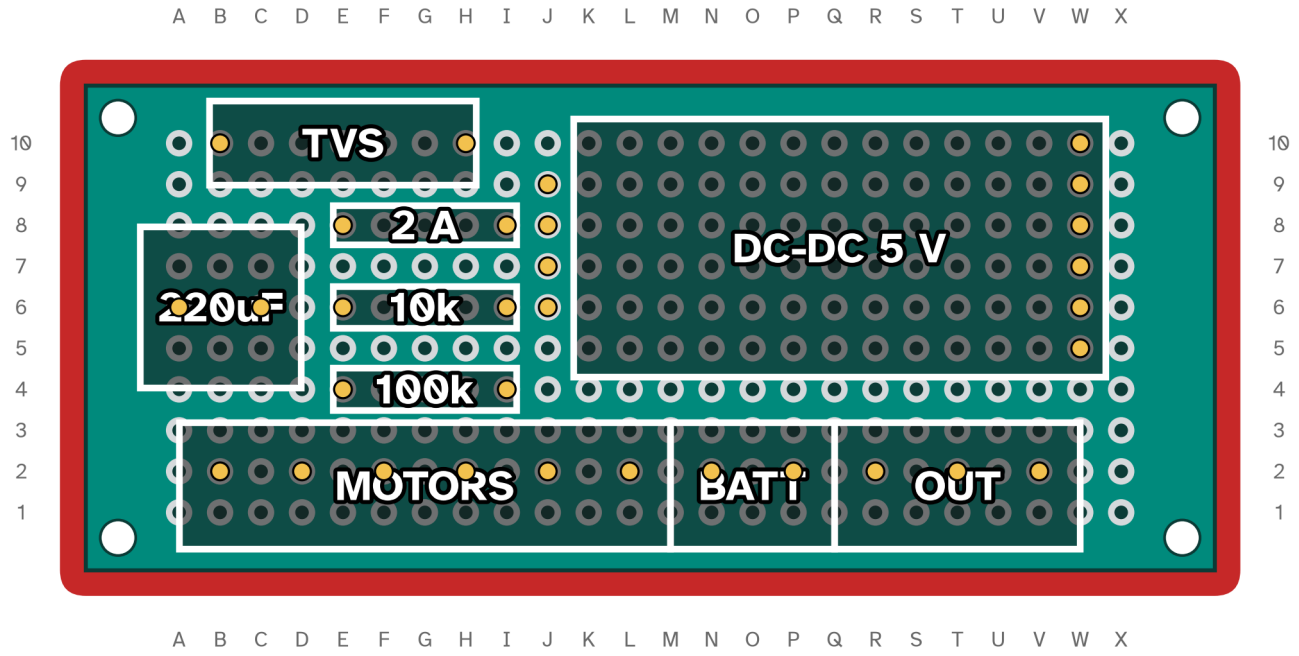
1x **Fuse 2 A, axial (green, 8 mm), soldered**
protects the DC-DC
holes: E8 - I8

1x **Resistor 100 kOhm, 1/4 W**
voltage divider for SENSE: + side
holes: E4 - I4

1x **Resistor 10 kOhm, 1/4 W**
voltage divider for SENSE: GND side
holes: E6 - I6

board z · where the parts go

Top side. Yellow dots = the pins.



Capacitor (polarized): the long leg (+) in C6, the stripe (-) side in A6. Turned round it can burst.

Terminal blocks: openings face the edge.

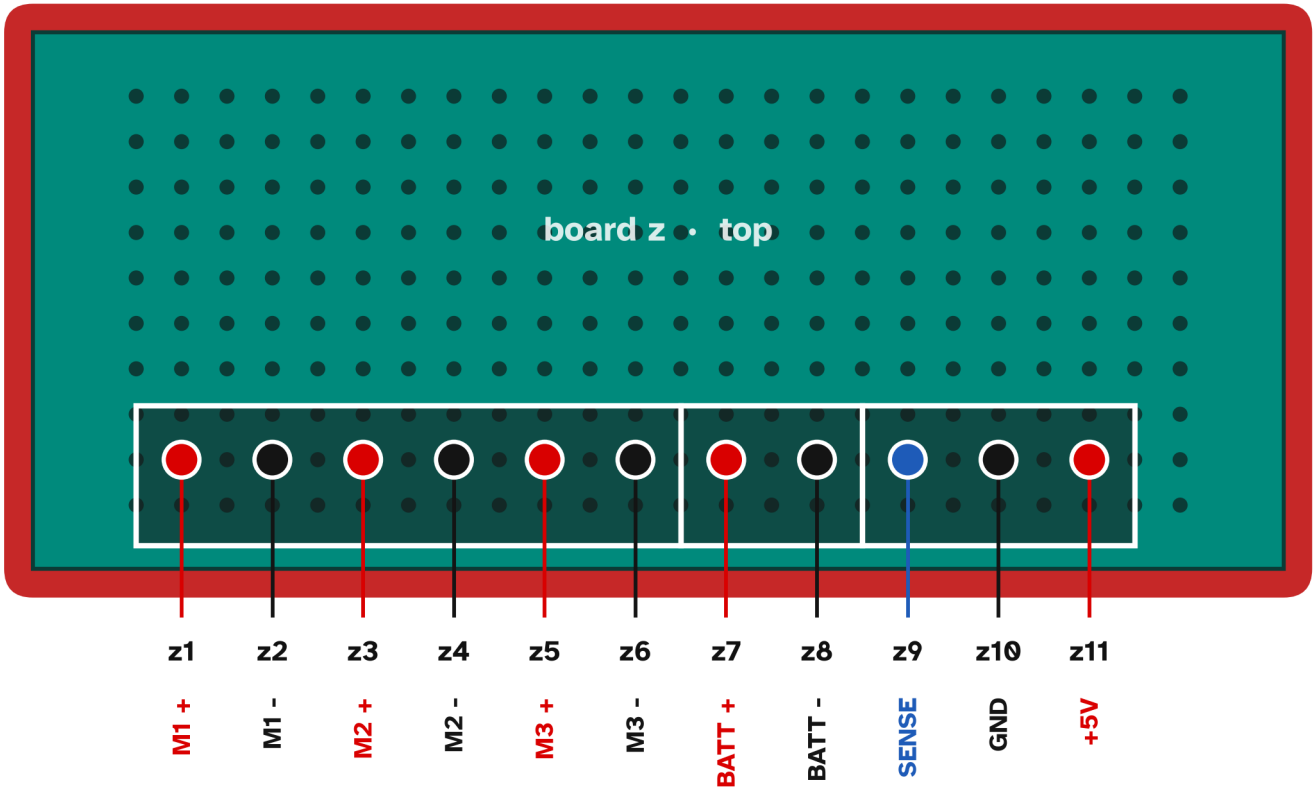
DC-DC: turned over, marking up. Its pins go into the female header in row W.

Tighten the 4 screw poles BEFORE it goes in: they face the board.

Screw poles: one wire each, straight down - GND into J9 and J8, VIN into J7 and J6.

board z · the terminals

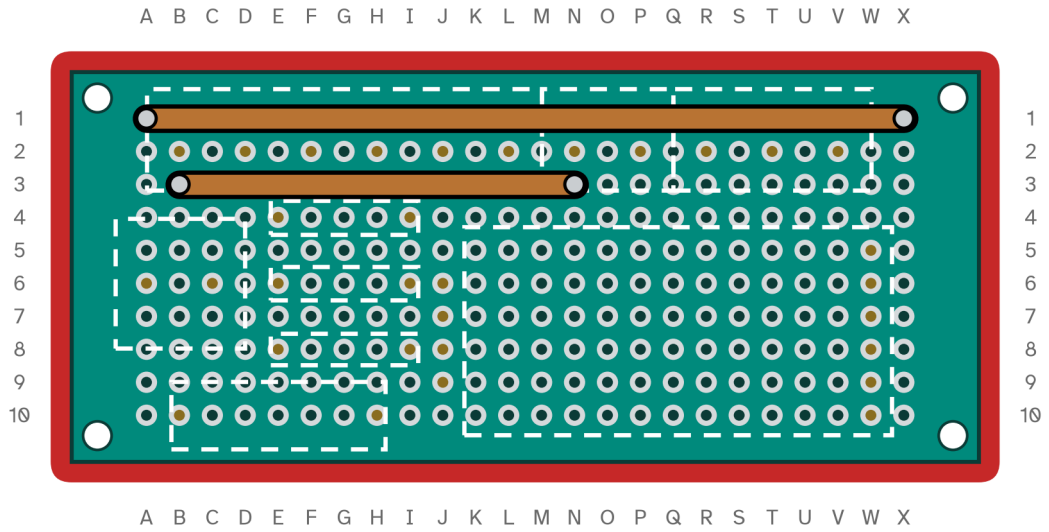
Each pole has a name: the board and a number. The Wiring guide uses the same names.



POLE	HOLE	JOB	THE WIRE GOES TO
z1	B2	**M1 +**	driver X
z2	D2	**M1 -**	driver X
z3	F2	**M2 +**	driver Z
z4	H2	**M2 -**	driver Z
z5	J2	**M3 +**	driver Y
z6	L2	**M3 -**	driver Y
z7	N2	**BATT +**	fuse
z8	P2	**BATT -**	battery connector -
z9	R2	**SENSE**	y3
z10	T2	**GND**	y2, x2
z11	V2	**+5V**	y1, x1

board z · Bare copper bus bars

Bottom side. Faded = already done.

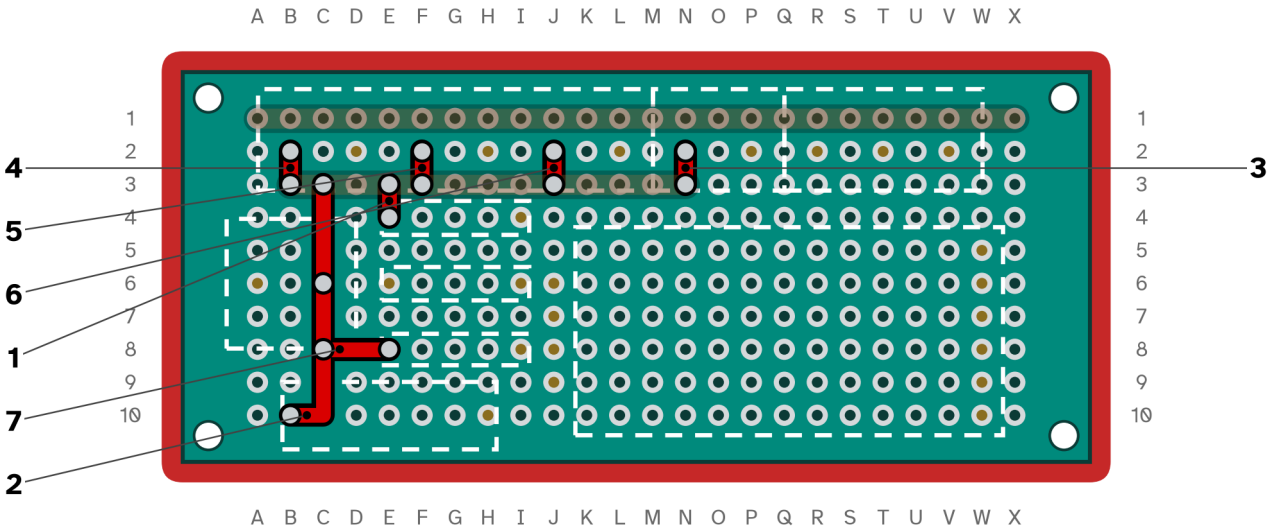


Bare 1.5 mm2 copper, filled with solder, soldered to every hole it passes.

B3 to N3	**30 mm**
A1 to X1	**58 mm**

board z · Red wires: +22V

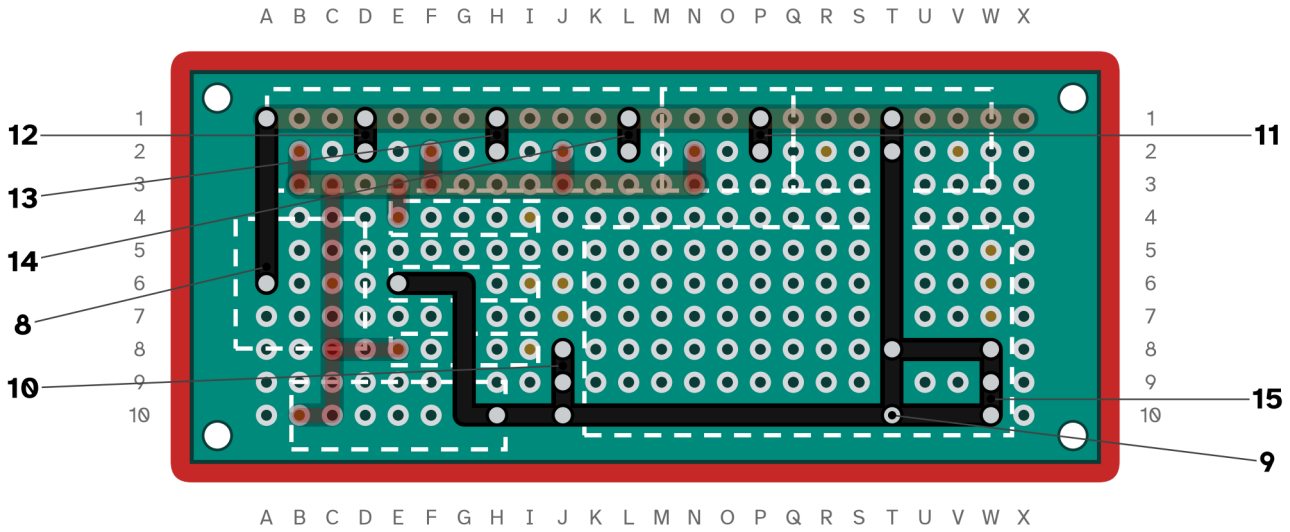
Bottom side. Faded = already done.



1	E4 100k a	-> E3 bus bar	**3 mm**
2	B10 TVS A	-> C3 bus bar	**20 mm**
3	N2 battery +	-> N3 bus bar	**3 mm**
4	B2 motors M1+	-> B3 bus bar	**3 mm**
5	F2 motors M2+	-> F3 bus bar	**3 mm**
6	J2 motors M3+	-> J3 bus bar	**3 mm**
7	E8 fuse in	-> C8 on wire 2	**5 mm**

board z · Black wires: GND

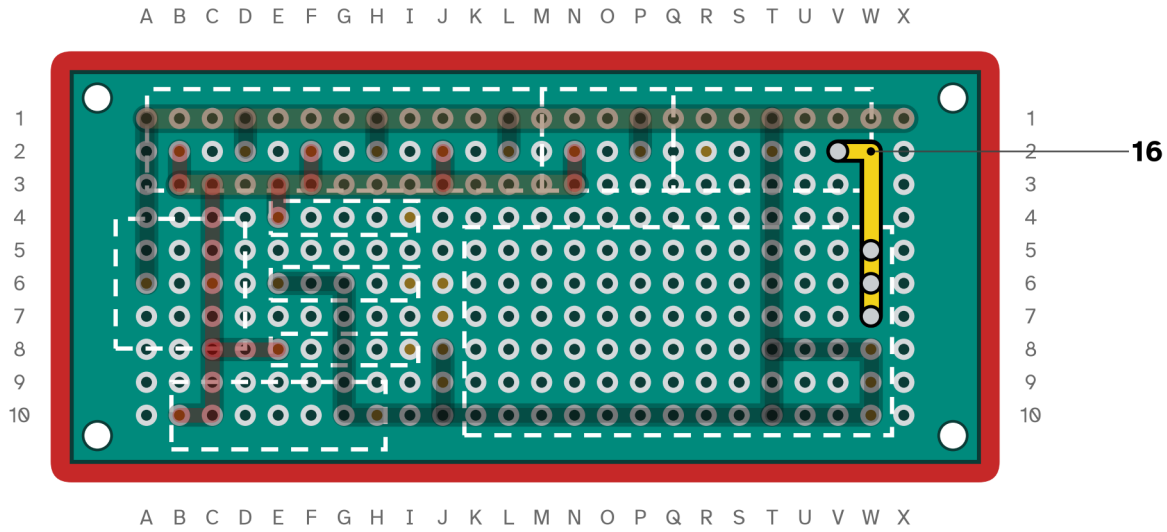
Bottom side. Faded = already done.



8	A6 cap. -	-> A1 bus bar	**13 mm**
9	E6 10k b	-> T1 bus bar	**71 mm**
10	J8 DC-DC pole GND2	-> J10 on wire 9	**5 mm**
11	P2 battery -	-> P1 bus bar	**3 mm**
12	D2 motors M1-	-> D1 bus bar	**3 mm**
13	H2 motors M2-	-> H1 bus bar	**3 mm**
14	L2 motors M3-	-> L1 bus bar	**3 mm**
15	T10 on wire 9	-> T8 on wire 9	**20 mm**

board z · Yellow wires: +5V

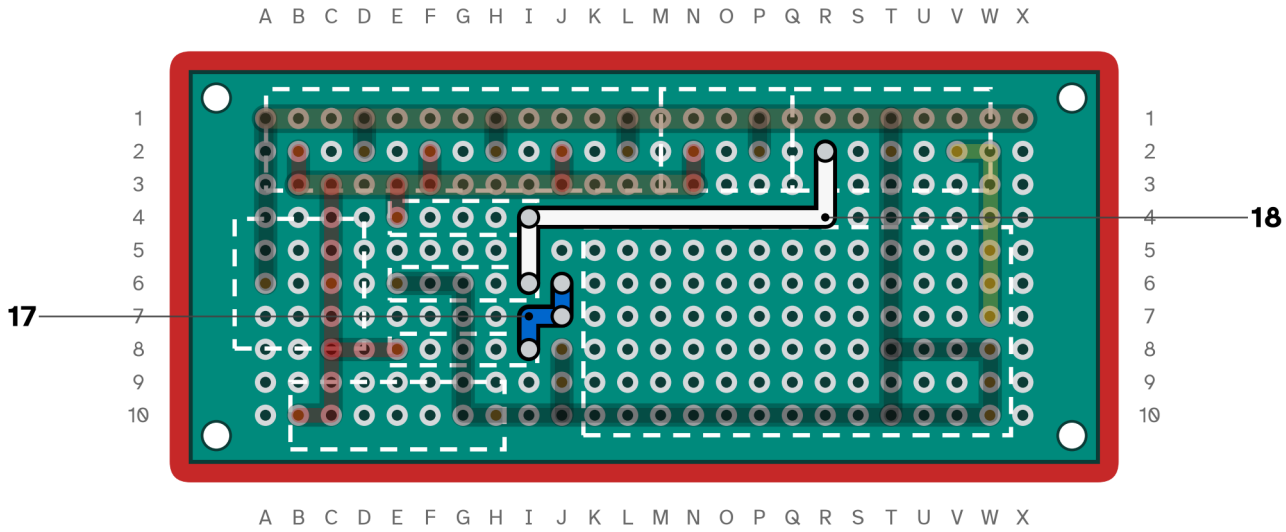
Bottom side. Faded = already done.



16	V2 output +	-> W7 DC-DC VO1	**15 mm**

board z · Signal wires

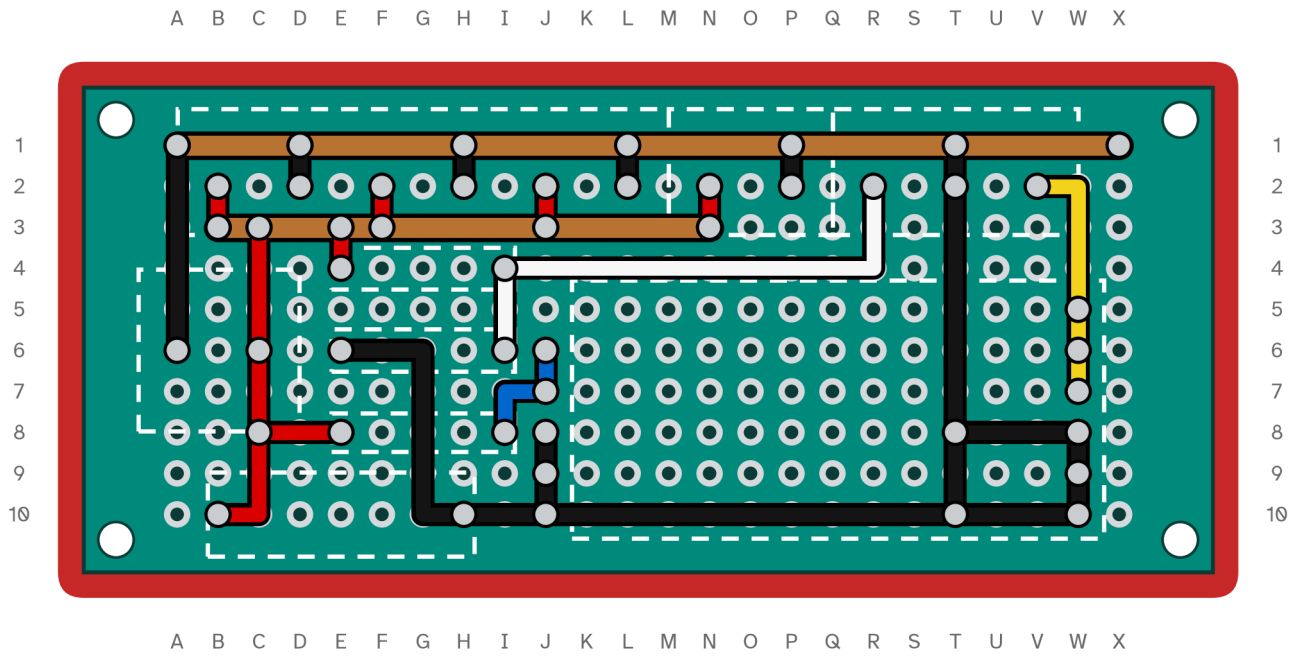
Bottom side. Faded = already done.

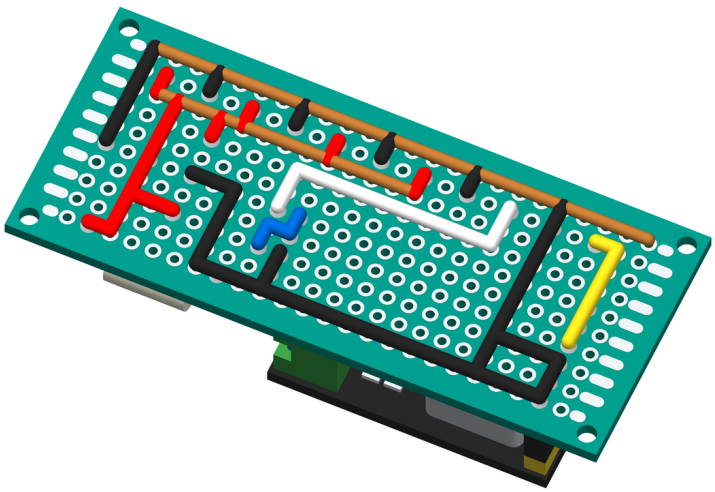


17	J6 DC-DC pole VIN2	-> I8 fuse out	**8 mm**
18	I6 10k a	-> R2 output S	**33 mm**

board z · done: like this

All the wires, bottom side.





board z · check: must beep

Multimeter on the beep setting. Board NOT connected.

+22V

N2 - B2 - F2 - J2 - B10 - C6 - E4 - E8

GND

P2 - D2 - H2 - L2 - H10 - A6 - E6 - T2 - W10 - W9 - W8 - J9 - J8

+22V after the fuse

I8 - J7 - J6

+5V

W7 - W6 - W5 - V2

SENSE

I4 - I6 - R2

Put one probe on the first hole, then touch each of the others: every one must beep.

board z · check: must NOT beep

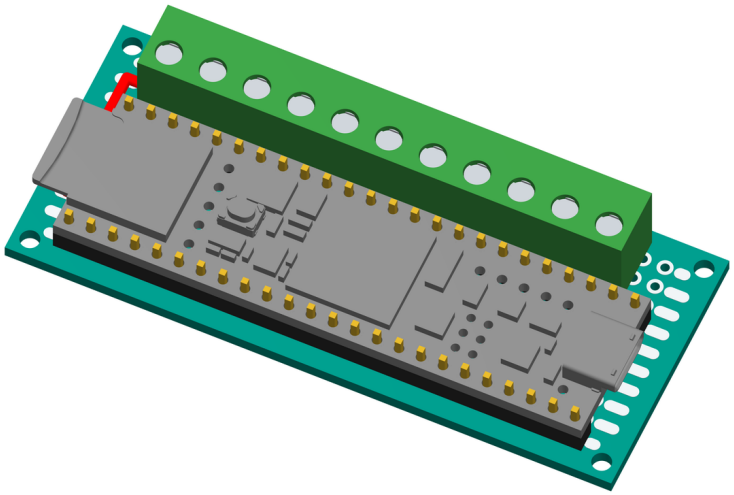
Multimeter on the beep setting. Board NOT connected.

- ⊗ +22V (N2) and GND (P2)
- ⊗ +22V (N2) and +5V (W7)
- ⊗ GND (P2) and +5V (W7)

A beep here is a short circuit: find it before you connect anything.

board y · MCU

The brain: the MCU board reads the IMU and the battery voltage and decides how the wheels must turn.



6 parts · 13 wires

Face y of the box. Find the letter Y on the box wall.

board y · the parts

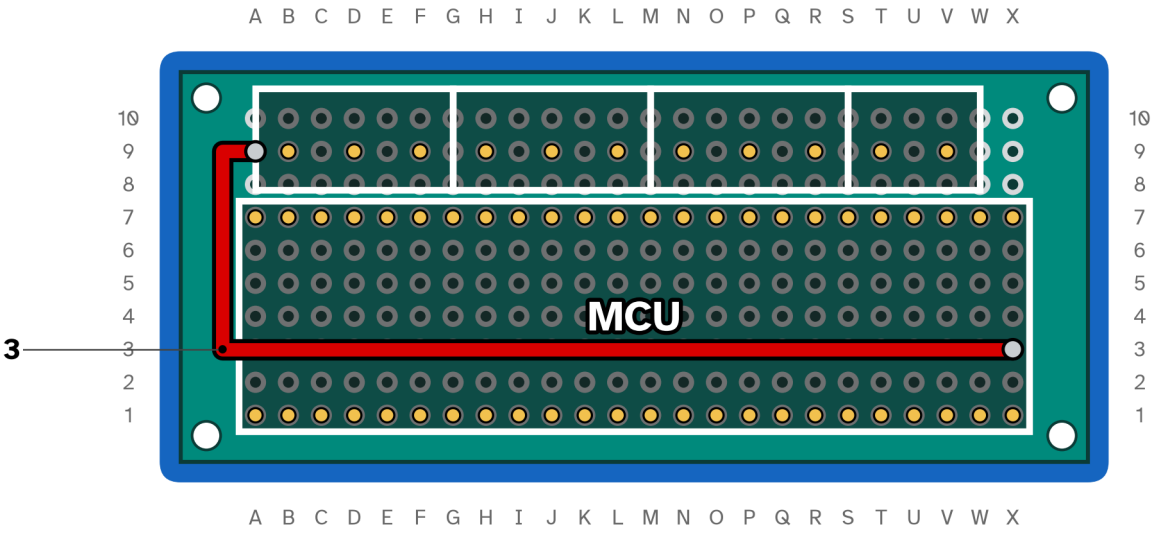
1x **MCU board on 2 pin headers (24 pins each)**
USB at row X
holes: columns 1 and 7, rows A ... X

4x **Terminal blocks: 3 x 5.08 mm 3-pole + 1 x 5.08 mm 2-pole**
one line of 11 poles
holes: column 9, rows B ... V

1x **Red insulated wire, thin: 24 AWG (about 1.2 mm with the insulation)**
+5V on the TOP side, under the MCU - 89 mm
holes: X3 - A9

board y · where the parts go

Top side. Yellow dots = the pins.



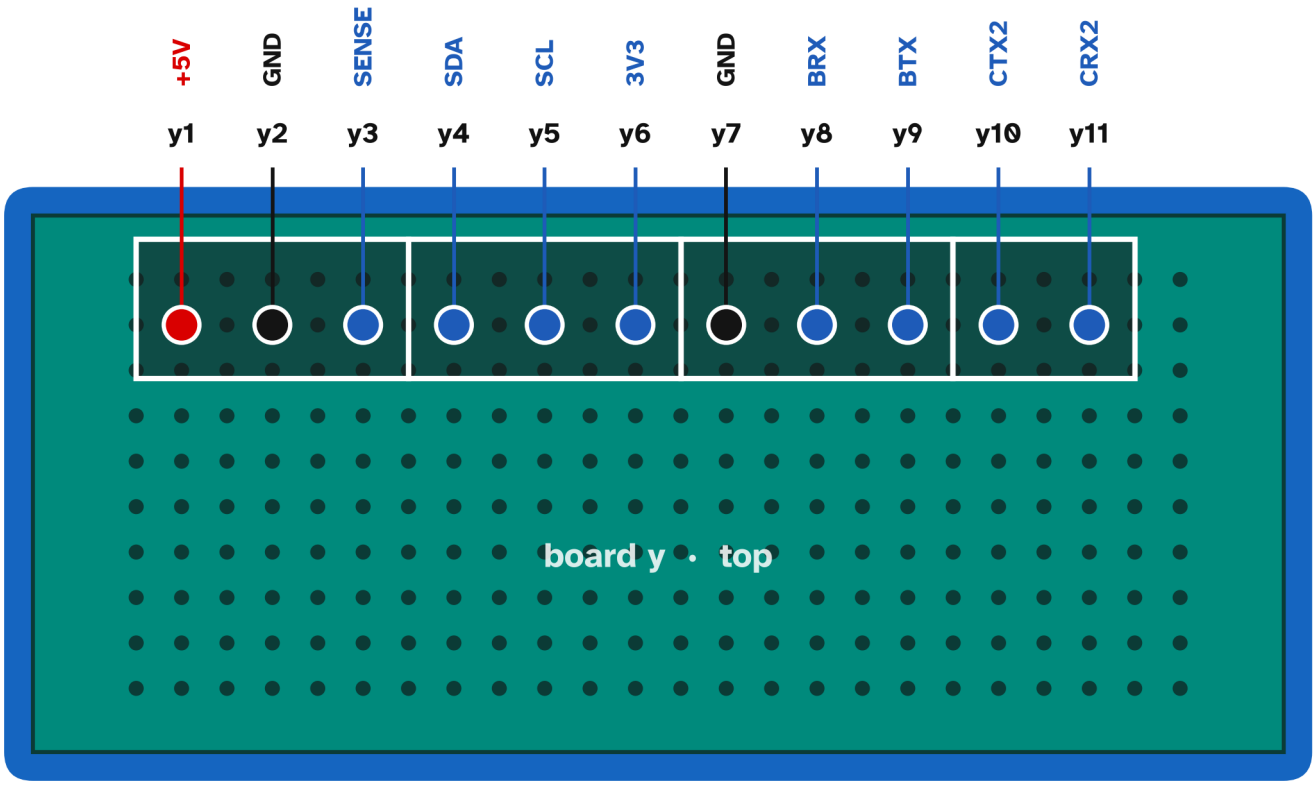
MCU: USB at row X, pins in columns 1 and 7.

Terminal blocks: openings face the edge, column 9.

Red wire 3 (+5V): the only insulated wire. Lay it BEFORE the MCU: from X3 along row 3, flat on the board (the MCU will sit over it), out past column A, down to A9. Its ends go down through the holes; you solder them from below, on wires 4 and 5.

board y · the terminals

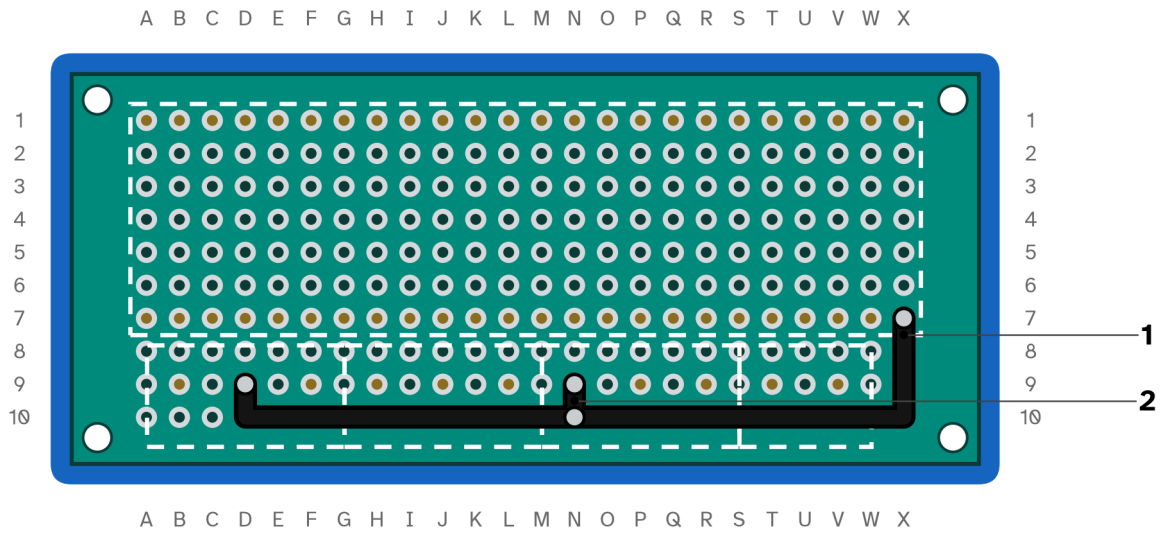
Each pole has a name: the board and a number. The Wiring guide uses the same names.



POLE	HOLE	JOB	THE WIRE GOES TO
y1	B9	**+5V**	z11
y2	D9	**GND**	z10
y3	F9	**SENSE**	z9
y4	H9	**SDA**	IMU
y5	J9	**SCL**	IMU
y6	L9	**3V3**	IMU
y7	N9	**GND**	IMU
y8	P9	**BRX**	x3
y9	R9	**BTX**	x4
y10	T9	**CTX2**	x6
y11	V9	**CRX2**	x7

board y · Black wires: GND

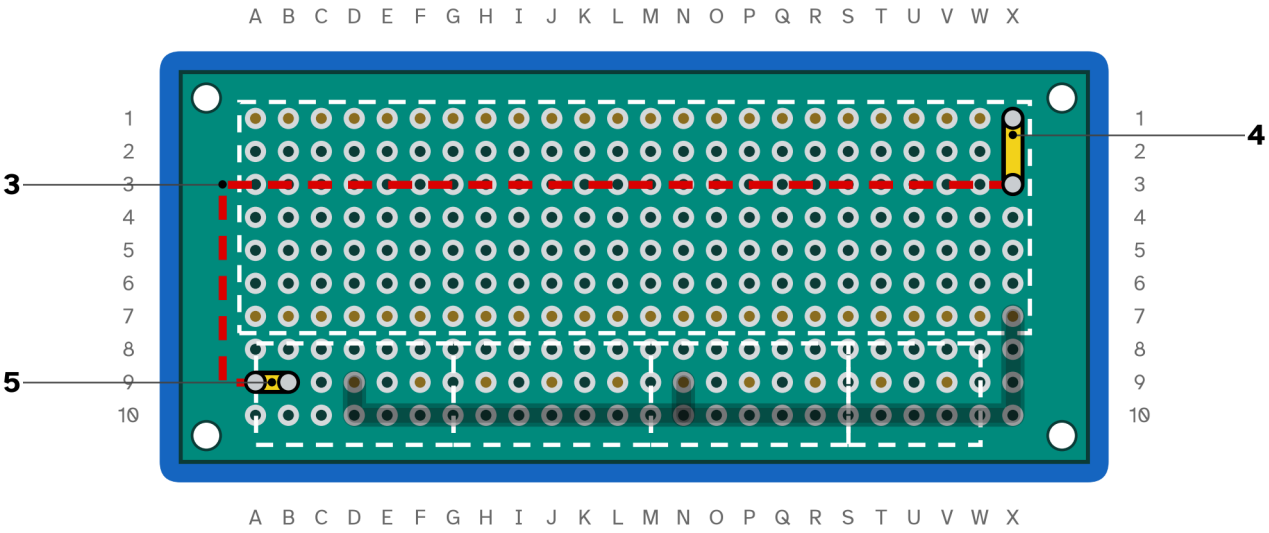
Bottom side. Faded = already done.



1	X7 MCU GND	-> D9 block GND	**61 mm**
2	N9 block GND_I	-> N10 on wire 1	**3 mm**

board y · Yellow wires: +5V

Bottom side. Faded = already done.

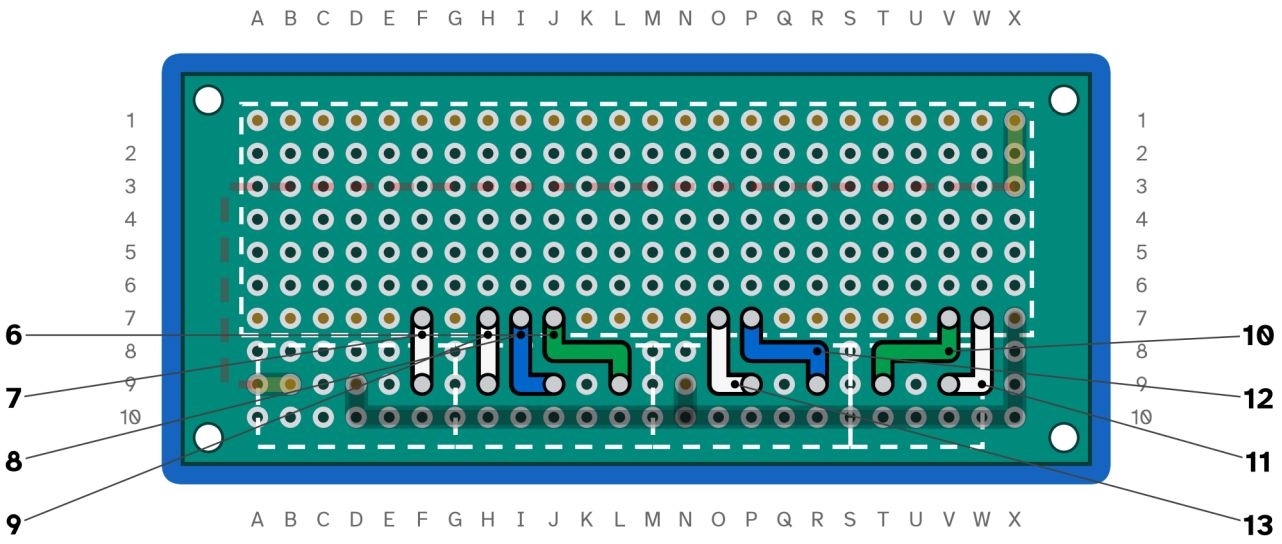


3	X3 top side	-> A9 top side	**89 mm**
4	X1 MCU VIN	-> X3 on wire 3	**5 mm**
5	A9 on wire 3	-> B9 block +5V	**3 mm**

Wire 3 is the red insulated wire on the TOP side (dashed here): laid before the MCU. Here you only solder its two ends, X3 and A9, together with the bare wires that meet there.

board y · Signal wires

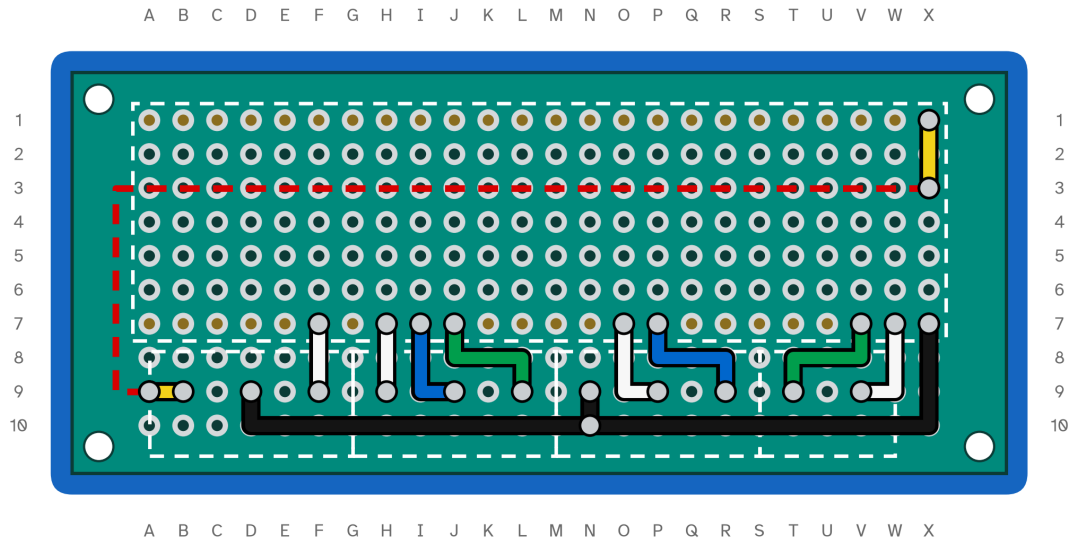
Bottom side. Faded = already done.

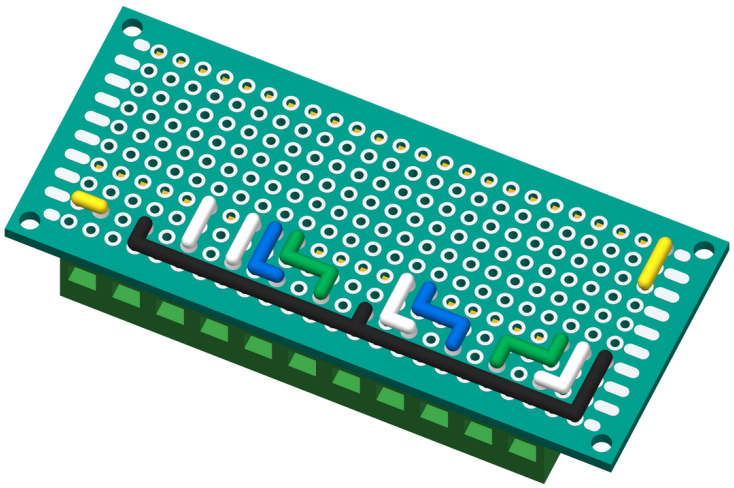


6	J7 MCU 3V3b	-> L9 block 3V3	10 mm
7	F7 MCU 27	-> F9 block SENSE	5 mm
8	I7 MCU 24	-> J9 block SCL	8 mm
9	H7 MCU 25	-> H9 block SDA	5 mm
10	T9 block CTX2	-> V7 MCU 1	10 mm
11	V9 block CRX2	-> W7 MCU 0	8 mm
12	P7 MCU 7	-> R9 block BTX	10 mm
13	O7 MCU 8	-> P9 block BRX	8 mm

board y · done: like this

All the wires, bottom side.





board y · check: must beep

Multimeter on the beep setting. Board NOT connected.

+5V

B9 - X1

GND

D9 - N9 - X7

3V3

J7 - L9

SENSE

F9 - F7

SCL2

J9 - I7

SDA2

H9 - H7

CTX2

V7 - T9

CRX2

W7 - V9

Wi-Fi board TX>RX2

R9 - P7

Wi-Fi board RX<TX2

P9 - O7

Put one probe on the first hole, then touch each of the others: every one must beep.

board y · check: must NOT beep

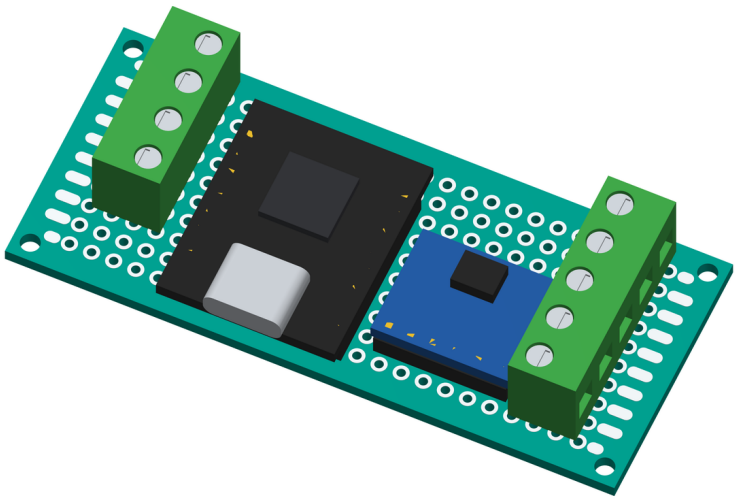
Multimeter on the beep setting. Board NOT connected.

- [ ] +5V (B9) and GND (D9)

A beep here is a short circuit: find it before you connect anything.

board x · WIRELESS + CAN

Talks to your laptop by radio (Wi-Fi board) and to the 3 motor controllers over the CAN bus (CAN board).



6 parts · 9 wires

Face x of the box. Find the letter X on the box wall.

board x · the parts

2x **Terminal blocks: 2 x 5.08 mm 2-pole**
+5V, GND, B-TX, B-RX
holes: C10 … C6

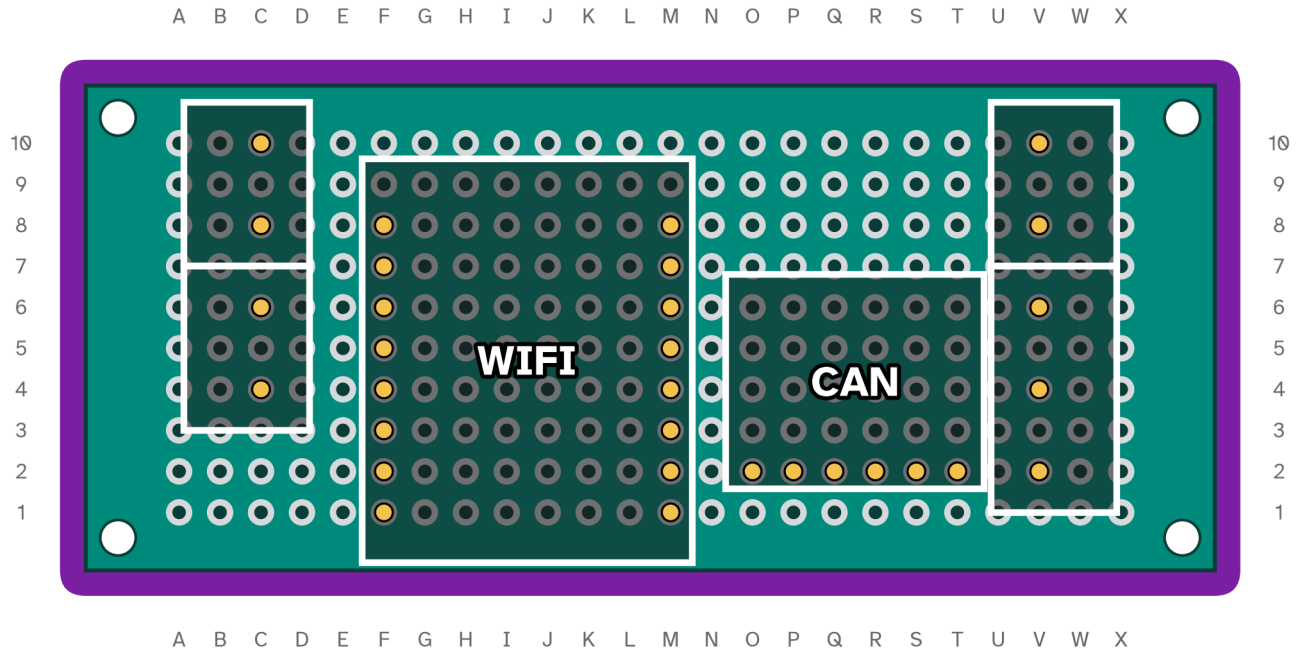
1x **Wi-Fi board, turned (on 2 x 8 pins)**
USB towards row 1; solder by the names on its board
holes: F1 … F8 and M1 … M8

1x **CAN transceiver board (on 6 pins)**
3V3 GND CTX CRX CANH CANL
holes: O2 … T2

2x **Terminal blocks: 5.08 mm 3-pole + 5.08 mm 2-pole**
CANL, CANH, CRX2, CTX2, GND
holes: V2 … V10

board x · where the parts go

Top side. Yellow dots = the pins.



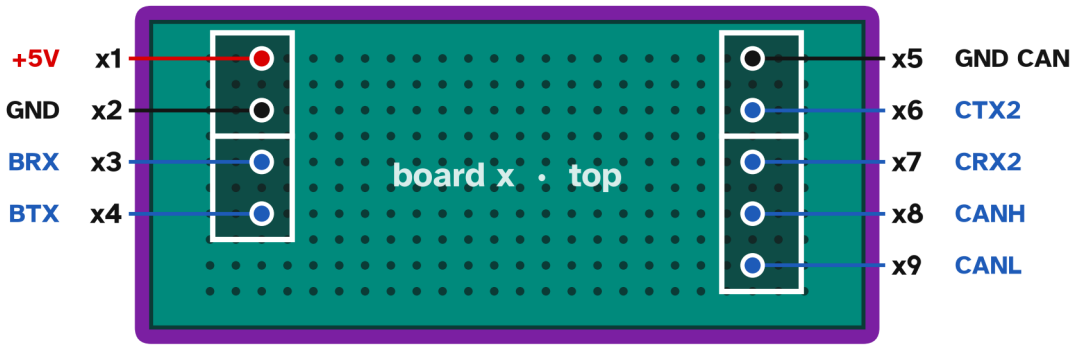
Wi-Fi board: turned, its USB towards row 1. Its BAT pin in F1, GND2 in M1 - check the names printed on it before you solder.

CAN board: 3V3 pin in O2 … CANL in T2, as printed on it.

Terminal blocks: one row in from the edge, openings out.

board x · the terminals

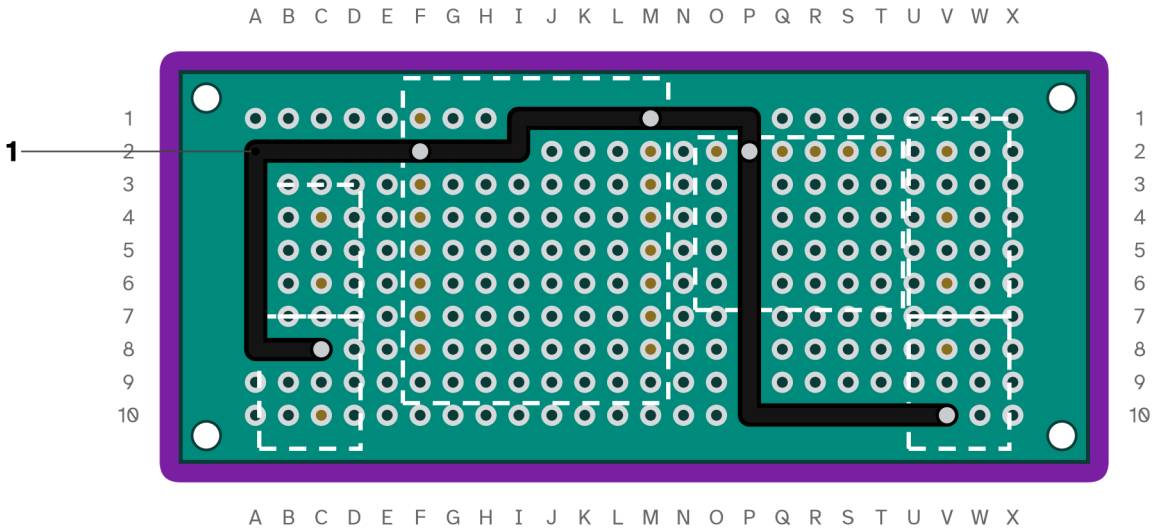
Each pole has a name: the board and a number. The Wiring guide uses the same names.



POLE	HOLE	JOB	THE WIRE GOES TO
x1	C10	**+5V**	z11
x2	C8	**GND**	z10
x3	C6	**BRX**	y8
x4	C4	**BTX**	y9
x5	V10	**GND CAN**	CAN 1 red
x6	V8	**CTX2**	y10
x7	V6	**CRX2**	y11
x8	V4	**CANH**	CAN 1 yellow
x9	V2	**CANL**	CAN 1 black

board x · Black wires: GND

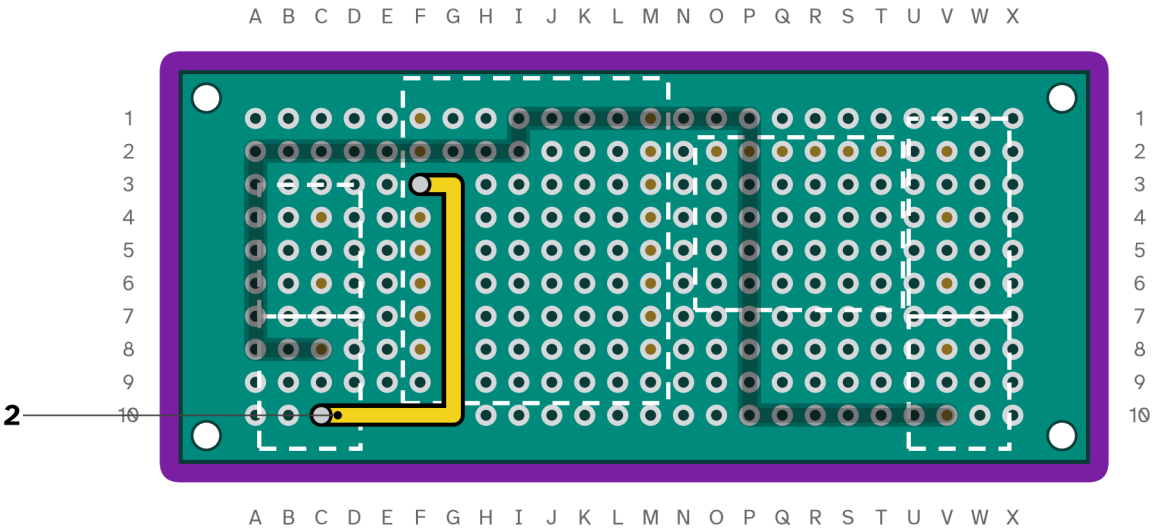
Bottom side. Faded = already done.



1	V10 block GNDC	-> C8 block GND	**99 mm**

board x · Yellow wires: +5V

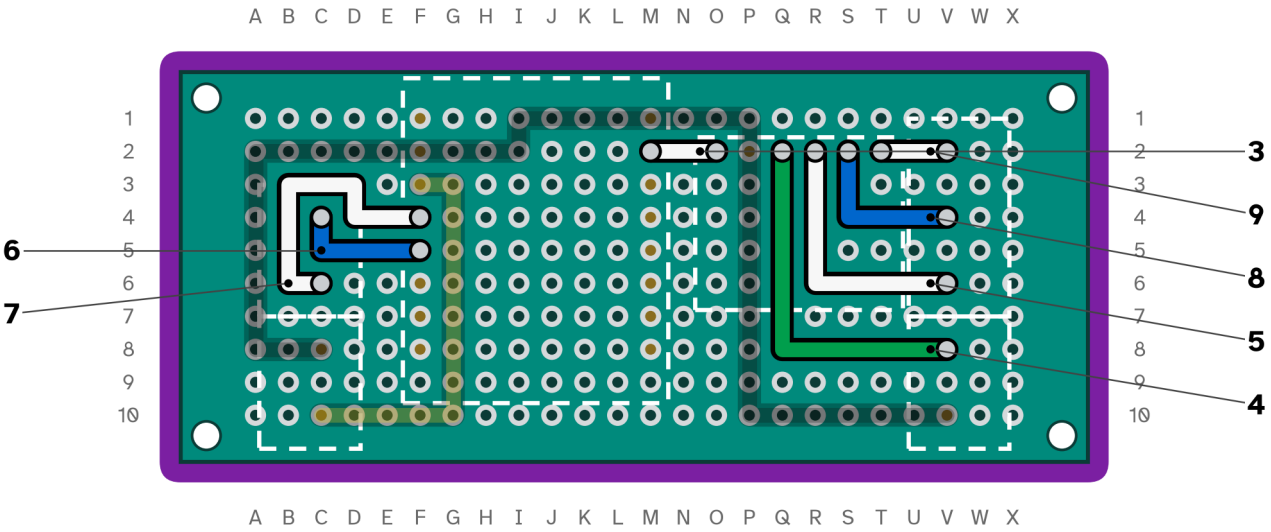
Bottom side. Faded = already done.



2	C10 block +5V	-> F3 Wi-Fi board VIN	**30 mm**

board x · Signal wires

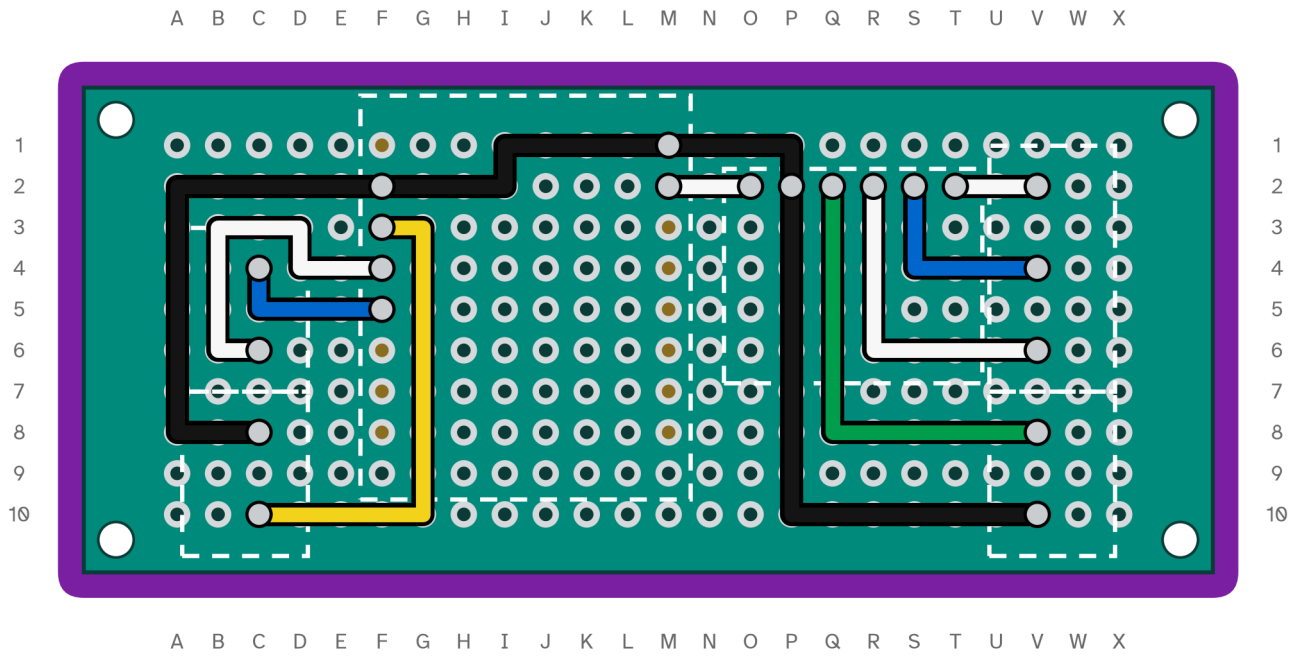
Bottom side. Faded = already done.

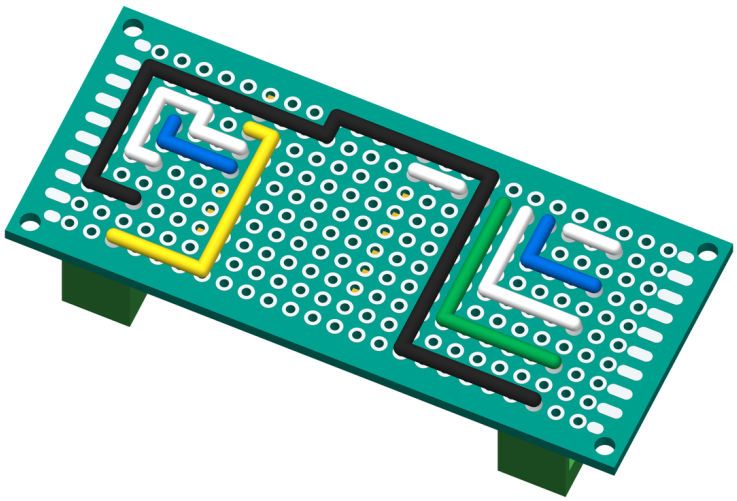


3	O2 CAN 3V3	-> M2 Wi-Fi board 3V3	**5 mm**
4	Q2 CAN CTX	-> V8 block CTX2	**28 mm**
5	R2 CAN CRX	-> V6 block CRX2	**20 mm**
6	F5 Wi-Fi board TX16	-> C4 block BTX	**10 mm**
7	C6 block BRX	-> F4 Wi-Fi board RX17	**23 mm**
8	S2 CAN CANH	-> V4 block CANH	**13 mm**
9	V2 block CANL	-> T2 CAN CANL	**5 mm**

board x · done: like this

All the wires, bottom side.





board x · check: must beep

Multimeter on the beep setting. Board NOT connected.

+5V

C10 - F3

GND

C8 - F2 - M1 - P2 - V10

3V3 Wi-Fi board

M2 - O2

CTX2

Q2 - V8

CRX2

R2 - V6

Wi-Fi board TX16

F5 - C4

Wi-Fi board RX17

F4 - C6

CANH

S2 - V4

CANL

T2 - V2

Put one probe on the first hole, then touch each of the others: every one must beep.

board x · check: must NOT beep

Multimeter on the beep setting. Board NOT connected.

- ⛔ +5V (C10) and GND (C8)

A beep here is a short circuit: find it before you connect anything.

Good to know

- **Not sure?**
  Look at the drawing again. Count the holes from the letter and the number.